Lora APRS - VHF APRS Bridge

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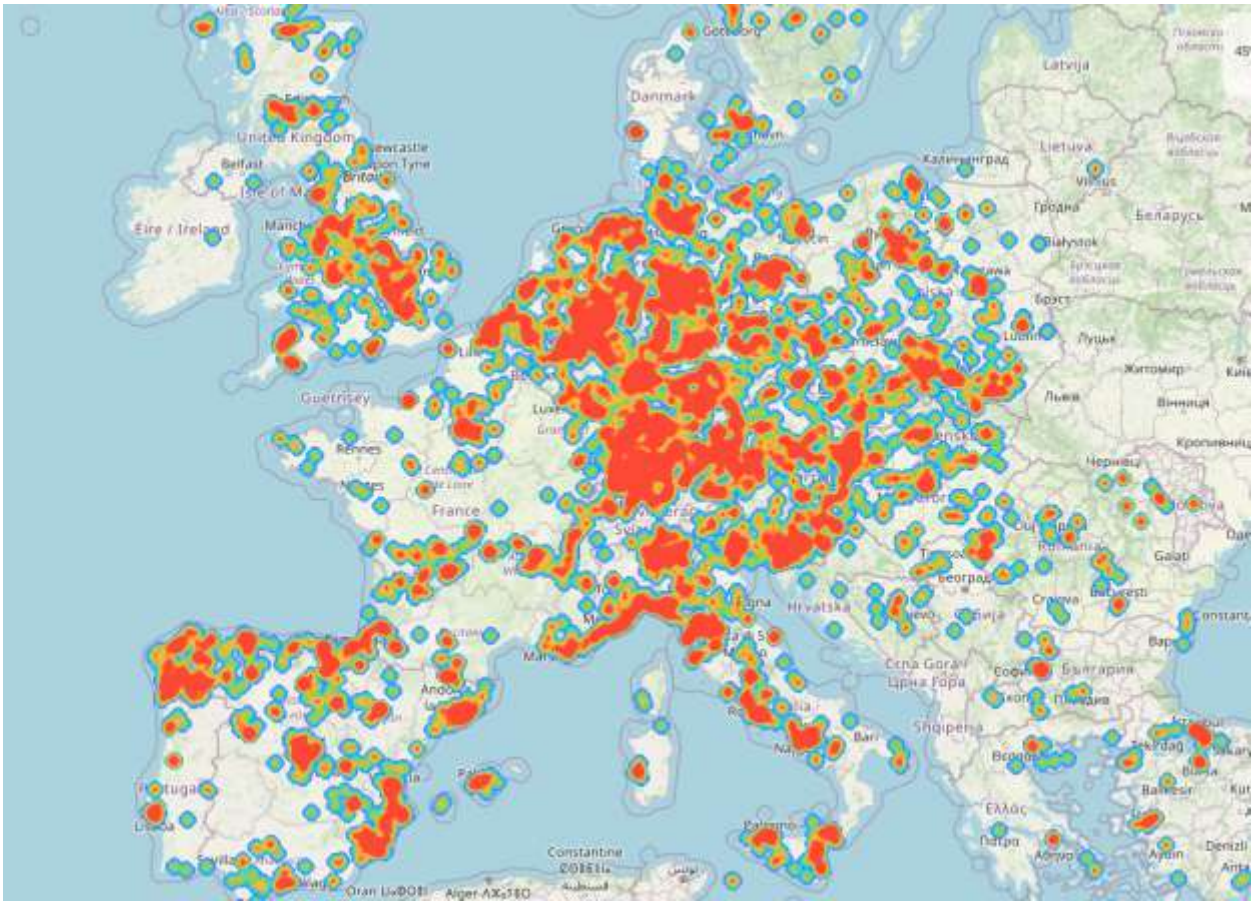
Background

LoRa is a radio technology that combines low-rate long range with low power by using spread spectrum. It is often used for Internet of Things and Meshtastic. Devices are very inexpensive. A typical board with LoRa, WiFi, Bluetooth, GPS receiver, USB port, battery charger, and OLED display is US $32.

LilyGo T-Beam

Recently there have been many new LoRa APRS projects gaining great popularity in Europe. They include trackers, digipeaters, and IGates. It is now starting to catch on in other parts of the world.

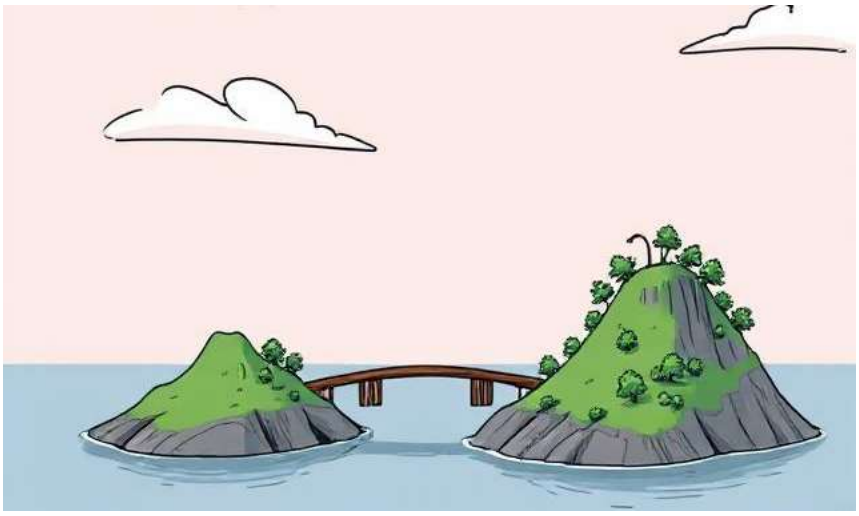


(from https://lora.ham-radio-op.net/)

There is a large chasm between these new projects and the existing VHF APRS network.

- Traditional 2 meter APRS users can't see LoRa APRS trackers.
- LoRa APRS users don't know what is happening on the local traditional APRS.
- "Messages" must pass through two IGates, and various servers, to get to someone across the street.

Geoffrey, F4FXL, made a big breakthrough by finding a way to provide bridging between LoRa APRS and traditional APRS. See Building a VHF / LoRa APRS Bridge .



LoRa APRS Traditional APRS

Here, is a simpler and more flexible approach.

You Can Play Along at Home -

You will need an IGate/Digipeater as described here: **iGate/digipeater firmware by CA2RXU**. Follow the instructions to connect it to your LAN.

It would be helpful to have a tracker as well so there is something to receive. **LoRa APRS Tracker Firmware** is known to be compatible. Be sure that they both use the same frequency and LoRa parameters. Temporarily set it to send a position report every 5 minutes to speed up testing.

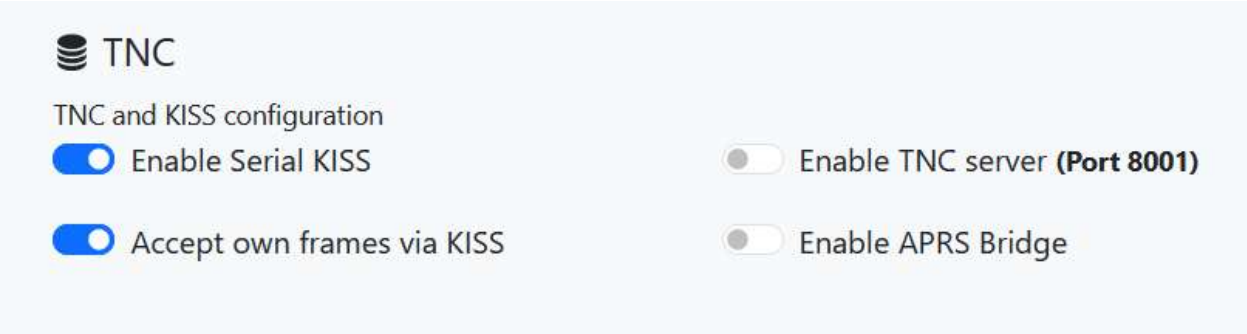
There are two different ways we can communicate with IGate/Digi node.

(1) As a KISS TNC over a serial port.
(2) As a KISS Network TNC over ethernet or WiFi.

(1) Access LoRa TNC by Serial Port

direwolf version 1.9 (currently the "dev" branch as this is being written) has the ability to access an external KISS TNC over a USB serial port.

The serial KISS option must be enabled in the LoRa TNC:

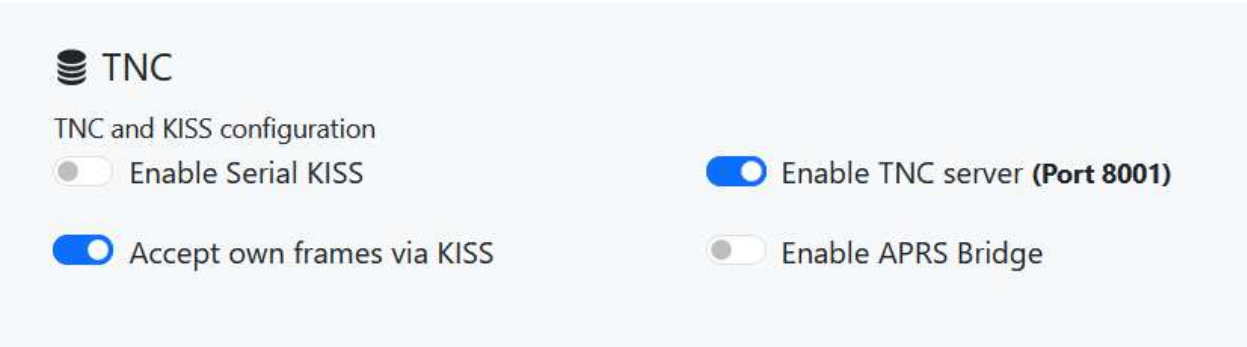


Next add a line to the configuration file to map the external TNC to a direwolf channel. Obviously, use the appropriate serial port name for your situation.

```
SCHANNEL 11 COM5 115200                # for Windows
SCHANNEL 11 /dev/ttyACM0 115200        # for Linux
```

(2) Access LoRa TNC over ethernet or WiFi

This was added in Release 1.8. Attach the LoRa device to your LAN and be sure to enable these options:



In my case, the LoRa Digi/IGate has an address of 192.168.1.238. When playing along at home, substitute your actual IP address and your own callsign.

My residential router also allows that address to accessed with the hostname iGATE-WB2OSZ-11 which would continue to work if the DHCP address ever changed. Your mileage may vary.

Add a line similar to one of these to your direwolf configuration file to map the network TNC to a direwolf channel:

```
NCHANNEL 11 192.168.1.238 8001
NCHANNEL 11 iGATE-WB2OSZ-11 8001
```

In Either Case (Serial Port or Network)

Do not enable the digipeater or IGate functionality. We will be using it as a simple KISS TNC.

First, we will run the “kissutil” application that is bundled in with direwolf. Wait until the tracker has a change to transmit a position.

For the network case, you should see something like this:

```
$ kissutil -h 192.168.1.238
[0] WB2OSZ-12>APLRT1,WIDE1-1:=/8wPx<K"SbV$G Bat=4.17V (100%)
[0] WB2OSZ-12>APLRT1,WIDE1-1:=/8wPg<K"BbU!G
```

For the serial port case, it is the same except the kissutil command uses the serial port name:

```
$ kissutil -p /dev/ttyACM0 -s 115200
```

WB2OSZ-12 is the name of the tracker.

APLRT1 identifies the application generating the packet.

The information part may look a little strange because it using the rarely used compressed format. If you have doubts, you can evaluate it with decode_aprs.

```
$ echo 'WB2OSZ-12>APLRT1,WIDE1-1:!/8wPq<^C,[;cQ' | decode_aprs

WB2OSZ-12>APLRT1,WIDE1-1:!/8wPq<^C,[;cQ
Position, Human, Ricardo, CA2RXU ESP32 LoRa Tracker
N 42 37.1398, W 070 30.3262, alt 39 m (129 ft)
```

See [Understanding APRS Packets]() for more information about demystifying the APRS packet formats.

Attach it to direwolf

Prepare a direwolf.conf file as shown below. ADEVICE, as shown, is for Windows; change appropriately for your operating system.

```
ADEVICE USB
#ADEVICE plughw:2,0

CHANNEL 0
MYCALL WB2OSZ-14

# Use NCHANNEL or SCHANNEL, depending on network or serial TNC.
NCHANNEL 11 192.168.1.238 8001
#SCHANNEL 11 COM5 115200              # for Windows
#SCHANNEL 11 /dev/ttyACM0 115200      # for Linux

OBEACON delay=0:15 every=10 via="WIDE1-1,WIDE2-1" sendto=11  symbol="dog"
objname="woofwoof" lat=42^37.17N long=071^20.82W comment="testing - no
comment"
```

Startup direwolf. Notice near the beginning, where channel properties are displayed:

```
Channel 0: 1200 baud, AFSK 1200 & 2200 Hz, A+, 44100 sample rate.
Channel 11: Network TNC 192.168.1.238 8001
```

Channel 0 is a normal radio channel using the internal TNC.

Channel 11 is now mapped to an external LoRaTNC. It can be treated like the normal channels for beaconing, digipeating, IGate, and packet filtering.

When we beacon to channel 11 (sendto=11), it is sent to the network TNC, i.e. the LoRa device.

```
[11>nt] WB2OSZ-14>APDW18,WIDE1-1,WIDE2-1:;woofwoof
*010000z4237.17N/07120.82Wptesting - no comment
```

If you have rsyslog running, you should see:

```
… …  WB2OSZ-11 CA2RXU_LoRa_iGate_1.3 TX / WB2OSZ-14>APDW18,WIDE1-1,WIDE2-
1:;woofwoof *010000z4237.17N/07120.82Wptesting - no comment
```

WB2OSZ-11 is the name of the LoRa Digi/IGate. "TX" indicates that it is transmitting.

It works in the other direction as well. We see the tracker (WB2OSZ-12) reception in the syslog.

```
… … WB2OSZ-11 CA2RXU_LoRa_iGate_1.3 RX / GPS / WB2OSZ-12 / APLRT1 / WIDE1-1 /
-87dBm / 11.50dB / -133Hz / 42.61898N / -71.34720E / 0.0km
```

Direwolf displays:

```
[11] WB2OSZ-12>APLRT1,WIDE1-1:!/8wQ?<K"$[<4Q
Position, Human, Ricardo, CA2RXU ESP32 LoRa Tracker
N 42 37.1334, W 071 20.8400, alt 43 m (141 ft)
```

Build a Bridge (really a cross-band digipeater)

Next, we want to forward packets from LoRa to VHF and vice versa. Add this to the configuration file.

```
DIGIPEAT 11 0 ^WIDE[3-7]-[1-7]$|^TEST$ ^WIDE[12]-[12]$

DIGIPEAT 0 11 ^WIDE[3-7]-[1-7]$|^TEST$ ^WIDE[12]-[12]$
```

It is easy to add sophisticated filtering to control what goes in each direction.

Example: Received on LoRa, forwarded to VHF APRS.

```
[11] WB2OSZ-12>APLRT1,WIDE1-1:!/8wQE<K"=[=gQ
Position, Human, Ricardo, CA2RXU ESP32 LoRa Tracker
N 42 37.1324, W 071 20.8321, alt 57 m (187 ft)

[0L] WB2OSZ-12>APLRT1,WB2OSZ-14*:!/8wQE<K"=[=gQ
```

- [11] means it was received over LoRa and mapped to channel 11.
- [0L] means it was cross-band digipeated onto VHF.

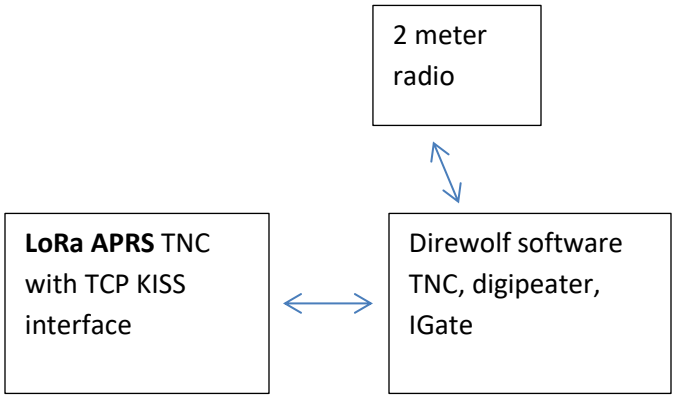
You can also perform the LoRa APRS digipeating to take advantage of the greater functionality provided by direwolf.

```
DIGIPEAT 11 11 ^WIDE[3-7]-[1-7]$|^TEST$ ^WIDE[12]-[12]$
```

Very flexible filtering is available to place any desired restrictions on what gets forwarded for each from/to channel combination.

```
FILTER  0  11  …etc.
FILTER  11 0   …etc.
FILTER  11 11  …etc.
```

See APRS-Digipeater-Algorithm.pdf for more information about digipeaters.



Let direwolf Perform the IGate Function for LoRa

The default is for all RF packets, received from all channels, to be forwarded to APRS-IS.

If, for some reason, you wanted to prevent LoRa packets from going to APRS-IS, you could add a filter:

```
FILTER 11 IG 0
```

This means:

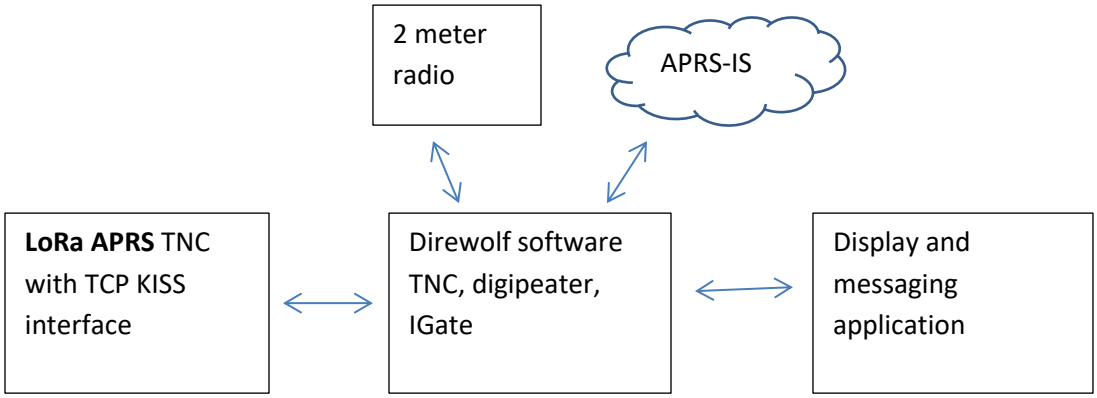
- If received on channel 11, the external TNC
- And going to APRS-IS.
- Condition 0 means false, don't pass it along.

In the other direction … to be continued …

See Successful APRS IGate Operation for more information about APRS IGates.

Add Your Favorite APRS Display Application

Configure your favorite APRS application (APRSISCE-32, YAAC, PinPoint APRS, Xastir, etc.) to use direwolf as the network TNC. Packets from both the VHF network(s) and the LoRa network(s) should all show up on the same map.



Other Uses

This new network/serial TNC feature is not limited to LoRa. It could be used with other TNC/modem types such as VARA, ARDOP, OpenTrac, CATS, DMR, D-Star, fldigi, M17, etc., and others not invented yet, as long as a network TCP or serial port KISS interface is available.